

BACKEND DEVELOPMENT

CURRICULUM · NODE.JS EDITION

Beginner to Advanced — Job-Ready Junior Backend Developer

INSTRUCTOR: *PAUL OLAGBOYE*

3

MONTHS

12

WEEKS

24

CLASSES

2–3 HRS

PER SESSION

2X CLASSES PER WEEK · 10–15 HRS SELF-PRACTICE RECOMMENDED

WHY THIS CURRICULUM

A relational-database-first stack — built for how companies actually hire

PostgreSQL is taught as the core database because most production backend roles run on SQL — not just NoSQL. The curriculum layers in the full toolkit employers expect from a junior hire.



Git & GitHub



API Development



Auth & Authorization



Database Design



Caching (Redis)



Testing



Security Best Practices



Deployment & DevOps



Real-World Projects

PROGRAM ROADMAP

Three months, four weeks each, two classes per week



MONTH 1

Backend Foundations

- Week 1 — Intro to Web Dev & JS Basics
- Week 2 — Advanced JavaScript (Async, ES6+)
- Week 3 — Node.js Fundamentals & Modules
- Week 4 — Git/GitHub & REST APIs



MONTH 2

Backend with Express

- Week 5 — Express Fundamentals & MVC
- Week 6 — Databases & Schema Design
- Week 7 — PostgreSQL + Prisma ORM
- Week 8 — Authentication, JWT & RBAC



MONTH 3

Advanced Backend

- Week 9 — File Uploads & Email Systems
- Week 10 — Pagination, Search & Redis Caching
- Week 11 — Security & Testing
- Week 12 — Deployment, CI/CD & Capstone

MONTH 1 · WEEKS 1–2

Backend Foundations

4 Classes

CLASS 1

Intro to Web Development

- Frontend vs Backend
- Client-Server Architecture
- HTTP/HTTPS, Request-Response
- Installing Node.js & VS Code

Practical: *Install dev environment, run first Node.js program*

CLASS 2

JavaScript Fundamentals

- Variables, Data Types, Operators
- Functions, Arrays, Objects
- Loops & Conditionals

Practical: *Build a simple calculator app*

CLASS 3

Advanced JavaScript

- Scope, Closures, Hoisting
- Callbacks, Promises, Async/Await

Practical: *Build an asynchronous file reader*

CLASS 4

ES6+ & Error Handling

- Destructuring, Spread Operators
- Modules
- Error Handling patterns

Practical: *Mini project using ES6 modules*

MONTH 1 · WEEKS 3–4

Node.js, Git & APIs

4 Classes

CLASS 5

Node.js Fundamentals

- Event Loop & Single Thread Model
- Core Modules & npm

Practical: *Build a CLI application*

CLASS 6

Built-In Node Modules

- fs, path, os, http in depth

Practical: *Build a simple HTTP server*

CLASS 7

Git & GitHub

- Git Basics & Branches
- Pull Requests & Collaboration

Practical: *Push first project to GitHub*

CLASS 8

REST APIs

- HTTP Methods & Status Codes
- JSON
- Testing with Postman

Practical: *Consume public APIs*

MONTH 2 · WEEKS 5–6

Express & Databases

4 Classes

CLASS 9

Express.js Fundamentals

- Routing & Middleware
- Request & Response cycle

Practical: *Build a simple API*

CLASS 10

CRUD & MVC

- CRUD Operations
- MVC Architecture

Practical: *Build a Student Management API*

CLASS 11

Databases Explained

- SQL vs NoSQL
- Database Design & ER Diagrams

Practical: *Design database schemas*

CLASS 12

Intro to PostgreSQL

- Tables & Relationships
- SQL Queries & Constraints

Practical: *Create a working database*

MONTH 2 · WEEKS 7–8

Auth & ORM Layer

4 Classes

CLASS 13

Node.js + PostgreSQL

- Connecting Node to PostgreSQL
- CRUD Queries
- Environment Variables

Practical: *Build an API backed by PostgreSQL*

CLASS 14

ORM with Prisma

- Migrations & Seeders
- Models

Practical: *Set up Prisma migrations*

CLASS 15

Authentication

- Auth vs Authorization
- Password Hashing
- JWT

Practical: *Build a user registration system*

CLASS 16

Authorization

- Role-Based Access Control
- Middleware Protection

Practical: *Secure API endpoints*

MONTH 3 · WEEKS 9–10

Files, Email & Scale

4 Classes

CLASS 17

File Upload & Email

- File Uploads & Image Handling
- Sending Emails

Practical: *Build profile picture upload*

CLASS 18

Cloud & Verification

- Cloud Storage Basics
- Email Verification

Practical: *Build a Forgot Password system*

CLASS 19

Querying at Scale

- Pagination & Filtering
- Searching & Sorting

Practical: *Build a Product API*

CLASS 20

Caching & Limits

- Caching with Redis
- Rate Limiting

Practical: *Optimize an API for performance*

MONTH 3 · WEEKS 11–12

Security & Ship It

4 Classes

CLASS 21

Security Best Practices

- Helmet & CORS
- Data Validation
- SQL Injection & XSS

Practical: *Secure an API end-to-end*

CLASS 22

Testing

- Unit & Integration Testing
- API Testing

Practical: *Write a test suite*

CLASS 23

Deployment & DevOps

- Env Vars & Logging
- CI/CD Basics
- Docker Introduction

Practical: *Deploy an API to production*

CLASS 24

Capstone Presentation

- Students present final projects
- Peer & instructor review

Practical: *Present the capstone build*

THE TOOLKIT

Every technology a junior backend developer is expected to know



RUNTIME

Node.js



FRAMEWORK

Express.js



DATABASE

PostgreSQL



ORM

Prisma



AUTHENTICATION

JWT



TESTING

Jest



API TESTING

Postman



CACHING

Redis



VERSION CONTROL

Git & GitHub



DEPLOYMENT

Docker

Plus Swagger/OpenAPI for documentation



CAPSTONE PROJECT

Class 24 — Students choose one production-style system to build & present

1



**E-Commerce
Backend**

2



**School
Management
System**

3



**Hospital
Management
System**

4



Social Media API

5



Job Portal API

Choose ONE · Deploy it live · Document the build

CAPSTONE REQUIREMENTS

What every final project must implement



Authentication & Authorization



PostgreSQL Database



REST API



File Upload



Search & Pagination



Email Notifications



Caching



Testing



Deployment



Documentation

FINAL PORTFOLIO

What every student walks away with



Authentication API

01



Blog API

02



E-Commerce Backend

03



Real-Time Chat Backend (Socket.IO)

04



Deployed Capstone Project

05

THE OUTCOME

Job-ready junior backend developer level in 3 months

Provided students commit to 10–15 hours of self-practice per week outside class — covering Git, REST APIs, authentication, PostgreSQL, caching, security, testing, and deployment along the way.

